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Quality dimensions, perspectives and practices

A mapping analysis

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Abstract *Reviews quality dimensions, perspectives and practices. An integrated framework is developed to analyze the relationships among quality dimensions, perspectives and practices. This framework incorporates two dimensions of the internal-external focus and objective-subjective measurement. It clarifies the concept of quality and helps practitioners to position their quality management practices. It explains how to resolve equivocality and reduce uncertainty in improving the quality of the products, ultimately to achieve competitive advantage.*

Introduction

For the past decade, firms have been challenged by competitors from Germany and Japan who produce better quality products at lower prices. To compete with the Germans and the Japanese, total quality management (TQM) has been widely adopted as an integrated management philosophy, the success of which was then proved by the pioneering companies such as Ford and Motorola and was marked in 1987 when the Malcolm Baldrige National Quality Award (MBQA) was established. The strategic importance of quality management practices has been acknowledged by both researchers and managers. Recent arguments about the tradeoff and cumulative models of competitive priorities make researchers and managers distinguish quality as the most basic competitive variable, without which other priorities such as cost, dependability and flexibility are impossible to realize (Ferdows *et al.*, 1990; Corbett and Wassenhove, 1993; Flynn and Flynn, 1996; Garvin, 1993; Noble, 1995). Although quality management practices have been studied in detail, the relationships between quality practices based on quality dimensions and perspectives warrant a further study (Dean and Bowen, 1994; Ghoshal and Bartlett, 1994; Godfray, 1993).

This article begins with a review on the quality dimensions, perspectives and practices in the literature. Then it develops a framework and five propositions to explore and analyze the relationships among quality dimensions, perspectives, and practices. Finally, a discussion on its implication is provided.

Review of quality dimensions, perspectives and practices

In the *Cambridge International Dictionary of English*, dimension is a measurement of something in a particular direction, especially, its height, length or width, or a part or feature. Perspective is the way of considering

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something or the method of representing in art. If something is put into perspective, it is compared with other things so that it can be accurately and fairly judged. Practice is something that is usually or regularly done, often as a habit, norm or custom.

Quality is a multi-dimensional construct (Garvin, 1988). Thus, quality can be seen differently from a particular angle (i.e. different perspectives). Basically, implicit multiple dimensions of quality lead to different perspectives about quality, which in return affect quality practices. The result of quality practices, the degree to which all dimensions of quality required by customers are met, creates competitive capability for the firm. The review framework is shown in Figure 1.

Each perspective makes different assumptions about how quality can be affected, accordingly it emphasizes different dimensions of quality. To address diverse quality dimensions equally effectively, managers have to recognize the limitations of each quality perspective since their quality perspective greatly limits their practices in which they might include some quality dimensions with the exclusion of other dimensions, while these dimensions may be more relevant to competitive advantages.

The multiple dimensions of quality

Eight principal quality dimensions are defined by Garvin (1988):

(1) Performance is a product's primary operating characteristic, such as acceleration, braking distance, steering, and handling of an automobile.

(2) Features are the "bells and whistles" of a product, such as power option and a tape or CD deck of a car.

(3) Reliability is the probability of a product's surviving over a specified period of time under stated conditions of use.

(4) Conformance is the degree to which physical and performance characteristics of a product match pre-established standards.

(5) Durability is the amount of use one gets from a product before it physically deteriorates or until replacement is preferable.

(6) Serviceability is the speed, courtesy, and competence of repair.

(7) Aesthetics is how a product looks, feels, sounds, tastes, or smells.

(8) Perceived quality is the subjective assessment of quality resulting from image, advertising, or brand names.

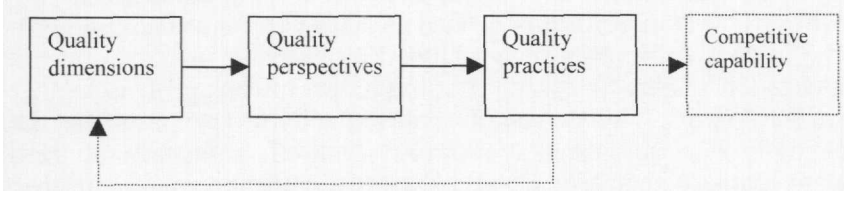


Figure 1.
The review framework of quality dimensions, perspectives and practices

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These are the basis for different perspectives of quality, although each perspective focuses on different quality dimensions and none of them captures all.

The perspectives of quality

The effective implementation and management of quality programs require consensus or cross-functional goals that are based on a shared understanding of quality definition. A list of different perspectives about quality is specified as follows.

Strategic quality is the strategic benefits that result from product and service quality (Garvin, 1988). Quality is a way to differentiate a product from its competitors (Porter, 1980). It potentially provides the producing firm with a sustainable competitive advantage that allows it to earn above average profits. A product or service that exceeds the quality of competing offerings can increase a firm's market share and improve customer perceptions of the product (Deming, 1986).

Transcendent quality, often used by consumers, is synonymous with superiority or excellence (Tuchman, 1980). In this sense, quality is both absolute and universally recognizable, a mark of uncompromising standards and high achievement. It is often loosely related to a comparison of features and characteristics of products and promulgated by marketing efforts aimed at developing quality as an image variable in the minds of consumers (Evans and Lindsay, 1996). Common examples of products attributed with images are Rolex watches, Mercedes-Benz and Cadillac automobiles.

Excellence is abstract and subjective; however, standards of excellence may vary considerably among managers and consumers. Hence, the transcendent definition is of little practical value to managers. It does not provide a means by which quality can be measured or assessed as a basis for decision making.

Product-based quality is a function of a specific, measurable variable. Differences in the quality reflect differences in the quantity of some product attributes, such as the number of stitches per inch on a shirt or the number of cylinders in an engine (Evans and Lindsay, 1996). It stems from the product design and operation. Taguchi and Clausing (1990) emphasize the extensive application of experimental design techniques in designing products. One of the problems with the definition is that it does not depend on external standards or reference; thus, the assessment of product attributes may vary considerably among individuals.

User-based quality is based on the presumption that quality is determined by what a customer wants. Individuals have different wants and needs, hence, different quality standards. It is defined as "fitness for intended use", or how well the product performs its intended function (Juran, 1951).

Value-based quality is based on value, the relationship of usefulness or satisfaction to price. This perspective sees a quality product is as useful as a competing product but is sold at a lower price. It offers greater usefulness or satisfaction at a comparable price (Broh, 1982). Competition makes businesses

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seek to satisfy customer's needs at a lower price. The value approach to quality incorporates a firm's goal of balancing product characteristics (the customer side of quality) with internal efficiencies (the operations side).

Manufacturing-based quality is defined as the desirable outcome of engineering and manufacturing practice or conformance to specifications. Specifications are targets and tolerances determined by designers of products and services (Crosby, 1979). Conformance to specifications is the key definition of quality since it provides a means of measuring quality. Specifications are meaningless if they do not reflect attributes that are deemed important to the consumer.

Total quality management practices

MBQA was instituted in 1987 to help stimulate US companies to improve quality and productivity while obtaining a competitive edge through increased profit. It also establishes the guidelines that can be used in evaluating quality improvement efforts and in learning how to manage for high quality. The criteria of MBQA and their relationships are shown in Figure 2 (The Malcolm Baldrige National Quality Award, The US Department of Commerce, National Quality Program, 1998).

Since the focus is on the implementation of quality management practices that are compatible with MBQA criteria, the following quality management practices categorized by MBQA are used for further discussion in this paper:

- management-based;
- employee-based;
- customer-based;
- information-based;
- product/process-based; and
- supplier-based.

Management-based quality practice. Management-based quality practice has two parts: top management support and strategic quality planning. Top management support has been identified as a major determinant of successful quality management, reasons being that top management support plays a vital role in projecting quality values, determining and enforcing the adoption of the values throughout an organization. In an organization, improving quality is a long-term competitive strategy and developing a quality culture is a lengthy process. Given the time factor, the organization should make a plan for achieving quality and integrate the quality improvement plan into the overall business plan. Although companies like to see immediate benefits from the quality improvement process from the very beginning, a long-term focus should be a greater objective.

Employee-based quality practice. Employee-based quality practice consists of four parts: employee training; involvement; empowerment; and reward and

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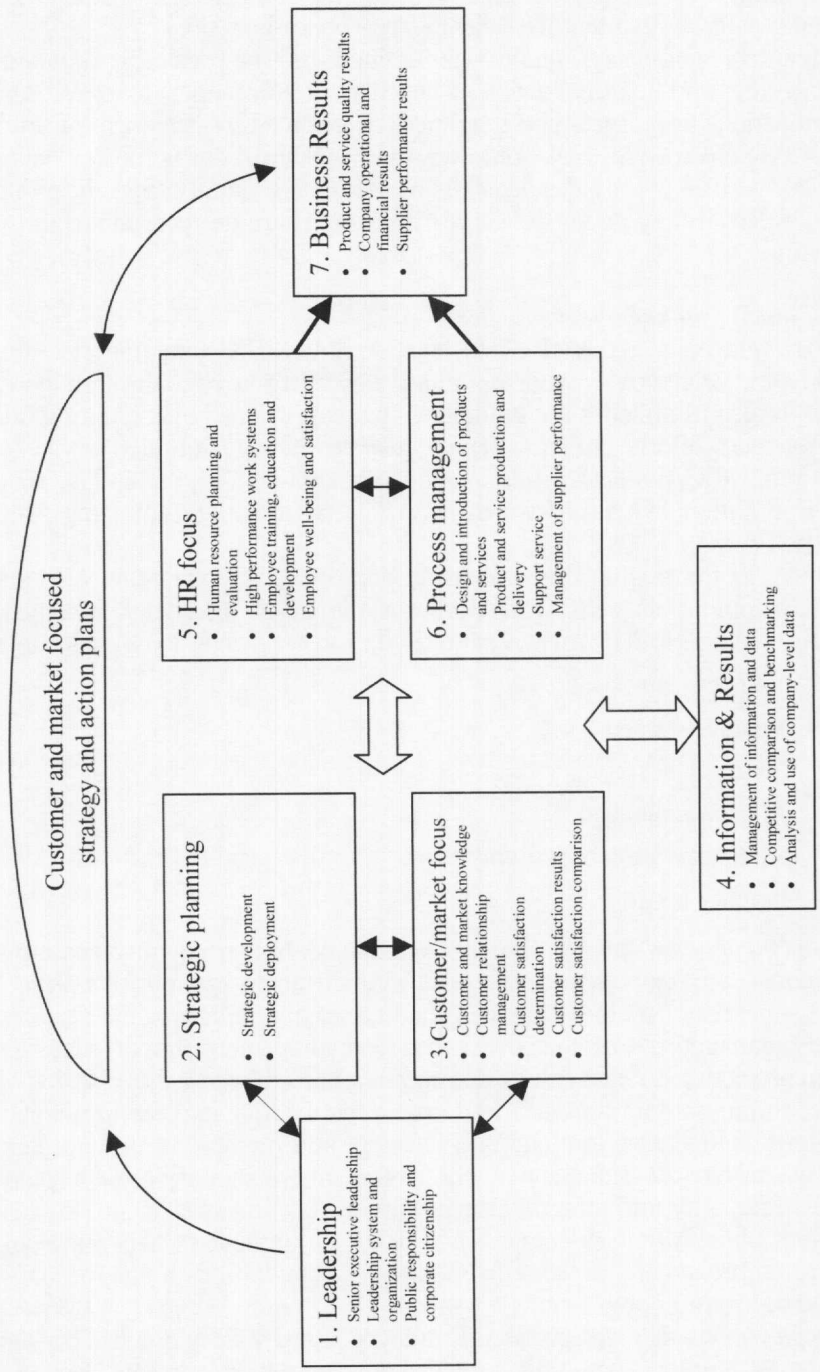


Figure 2.
Baldrige criteria for performance excellence framework: a system perspective

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recognition. Employee training in quality-related concepts and tools are prerequisites for the effectiveness of quality improvement activities. More than any other factors, human resources are the most important in determining the long-term success of a quality management firm. In addition, the knowledge of basic quality concepts and principles facilitates the understanding of quality relevant issues and provides a common language for cross-functional teams to problem solving. The basic idea behind involvement is that no others but employees control the work and participate in the business of an organization such as problem-solving and decision-making. Empowerment motivates workers to do their work better. Since people are the creative source for new ideas and innovation, more satisfaction can be perceived. Recognition and reward play three important parts in a TQM environment: an indicator of performance; a feedback tool; and a way to show appreciation for effort by the organization. Rewards are used to improve performance in teamwork, TQM tools, quality problem-solving, and interaction with internal and external customers. A reward system reflects the corporate philosophy and cements employee commitment to organizational values. It is a public statement about the priorities of the organization, consisting of quality, customer satisfaction, and continuous improvement.

Information-based quality practice. Another key element of quality management success is information (Godfray, 1993). Three elements of information-based quality practices form a sound foundation: availability; usage; and benchmarking. However, information-based quality practices are almost an untapped area. Few researchers look at the data and information needs of a company engaged in serious TQM efforts. The traditional random management of quality information systems with inaccurate, incomplete and outdated information is far from adequate for the competitiveness of the company. In effect, the availability of exact information on quality is a prerequisite for effective and efficient quality management practices. It is a solid foundation for the development of critical competitive capabilities (Schlange, 1991; Adam and Swamidass, 1989). In essence, the use of quality information can aid manufacturing companies to achieve effective and efficient quality. For example, Xerox transformed quality data into useful information which was applied to drive quality improvement actions such as improving the next generation of products and business process, reducing cycle-times, furthering distribution and field service, improving understanding of customer needs, and upgrading products and services designs. Benchmarking helps an organization identify, understand and adapt outstanding practices from leading competitors in the same industry and those in the other industries who use similar processes to improve performance. When comparing with the "very best", the organization can make improvements in products and processes.

Product/process-based quality practice. Product/process-based quality practice comprises two components: product design; and process design and control. The quality assurance of product design affects internal quality performance and competitive capabilities through its effect on the

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manufacturability, complexity, reliability, features, and serviceability of the products. Of critical importance is the assurance of incorporating customer-desired product features at the early stage because making changes in product quality or customer value at the production stage disrupts the efficiency and productivity of the manufacturing process. The quality assurance of work process design is believed to be a crucial practice with the flexibility, reliability, and maintainability of a process. Statistical process control becomes the core for both quality improvement and quality maintenance in minimizing production cost and attaining consistency of products and services.

Customer-based quality practice. Customer-based quality practice contains customer orientation and closeness. In a customer-oriented organization, total customer satisfaction becomes the goal of the entire organization. TQM is organized for the purpose of customer satisfaction – what customers expect is the most important job the company has (Wheaton and Weimerskirch, 1989). Furthermore, customers are not only sources of market information, but also providers of key resources including technology, knowledge, skills and other essential services necessary for the competitiveness of the manufacturers. They help manufacturers to make decisions.

Supplier-based quality practice. Supplier-based quality practice provides a means to increasing the likelihood of an organization to have reliable suppliers who are willing to work towards the company's goals of achieving quality excellence. Because the quality of products depends to a large extent on the quality of the components and because a company needs full cooperation from its suppliers to design and develop new products with higher reliability in shorter time, the relationship with suppliers becomes critical for a company's success.

A framework of analysis

Garvin (1986) explains in an exploratory study the differences between Japanese and US manufacturers in the process to gain competitive quality. In Japan, quality problems are attributed to external factors such as poor materials or purchased parts; while in the USA, the problems stem from internal sources such as workforce, maintenance or adjustment of equipment, or processes (Nakane, 1986). The Japanese strategic objective is to produce high quality products; whereas US managers tend to subordinate quality objectives to the drive to meet production schedules.

The basic approach of TQM puts customer satisfaction as the driving force behind quality practices. Thereby, the internal and external customer focus of the product or service becomes the main consideration for determining standards and measuring performance (Raturi and McCutcheon, 1990).

Another important characteristic of the TQM approach is continuous improvement. In general, TQM organization is dynamic and is constantly striving to improve. So measurement of information becomes vital. The principal goal for everyone in the organization is to make decisions that can meet or exceed customer expectations while using the organization's resources

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productively. In order to make such decisions, employees need information about customers, products and services, processes, suppliers, competitors, and employees themselves (Evans and Lindsay, 1996).

Information derives from analysis of data. Data, in turn, come from measurement. Thus, achieving quality excellence requires a framework of measurement. Measurement is the act of determining whether an object possesses a certain quality characteristic (usually subjective), or quantifying the amount of a quality characteristic. Measurement allows assessment of the degree of conformance to a specification, which, in turn, enables managers and front-line workers to know how close they are to their targets so as to set improvement priorities. Measurement provides a scorecard for business performance and makes accomplishments visible to everyone. Based on the previous literature, a framework capturing the relationship among production cycles, quality perspectives and quality dimensions is shown in Figure 3.

The framework is developed using the two dimensions of internal-external focus and subjective-objective measurement. These two dimensions echo the information processing theory proposed by Daft and Lengel (1986). They think that the organization faces two kinds of information requirements to reduce the equivocality and uncertainty. Equivocality can be resolved with more communication to clearly define the problem (i.e. a single, uniform interpretation) while the uncertainty can be reduced with more information added to the defined problems. For quality management, to resolve the

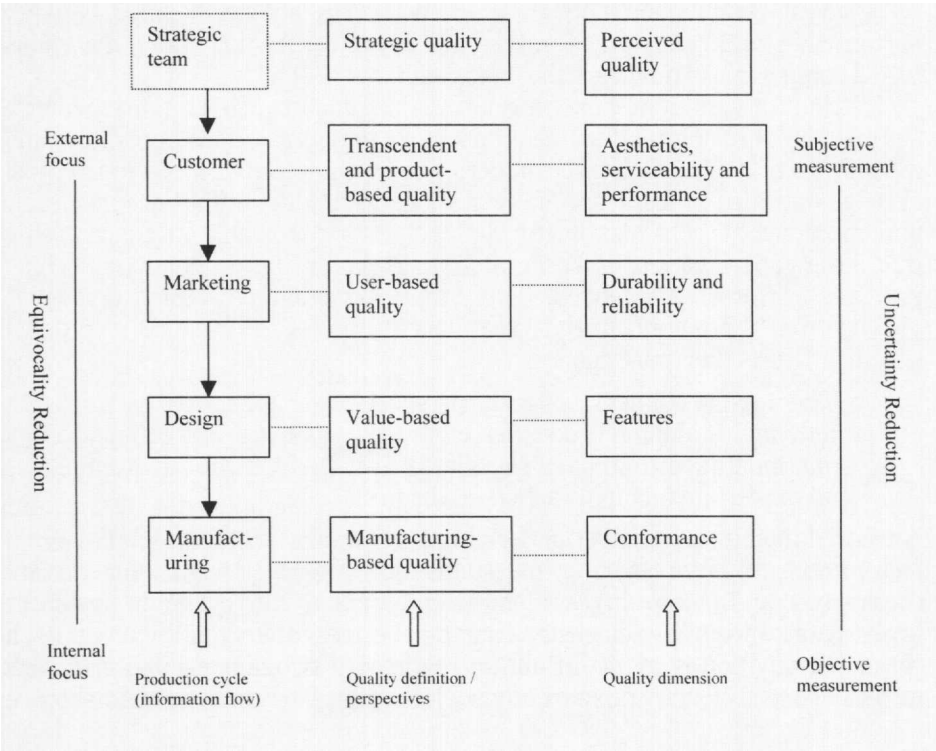


Figure 3.
The relationship between quality perspectives, dimensions and production cycle

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equivocal problems, managers have to communicate effectively to come up with the unified interpretation about quality definitions and practices; to reduce the uncertainty problems, the organization has to obtain more information about existing processes and procedures.

P1: The quality dimensions emphasized by different individuals along a production cycle (i.e. customers, marketing, design, and manufacturing) have significant relationships with their perspectives of quality.

Although each individual throughout a production cycle is important to the product quality, how quality is defined depends on one's position in the system, a designer, a manufacturer, a distributor, or a customer. The customer is the driving force of the production of goods and services. Customers generally view quality either from a transcendent perspective or from a product-based perspective. Goods and services are produced to meet customers' needs with dimensions such as aesthetics, serviceability and performance. It is the role of the marketing function to determine these needs. A product that meets customer needs can be called "quality". Hence, the user-based perspective of quality such as durability and reliability is meaningful to marketing people.

Customer needs should be translated into product and process specifications by the manufacturer. "Translating" is the main task of R&D, product design, and engineering. Product specifications include such features as size, form, finish, taste, dimensions, tolerances, materials, operational characteristics, and safety. Process specifications include types of equipment, tools, and facilities to be used in production. Thus product designers should balance between performance and cost to meet marketing objective. At this stage, the value-based perspective of quality is the most useful.

All kinds of variations can occur during manufacturing operations: machine settings may fall out of adjustment; machines may break down; operators and assemblers may make mistakes; materials may be defective. Even in the most closely controlled process, specific variations in product output are inevitable and unpredictable. The manufacturing function is responsible for guaranteeing that design specifications are accurately followed during production and final products perform as intended. Thus, for production personnel, quality is viewed from the manufacturing-based perspective. Conformance to product specifications is their major concern.

P2: The quality dimensions and perspectives emphasized by different individuals along a production cycle (i.e. customers, marketing, design, and manufacturing) have significant relationships with their emphasis on the certain quality practices in the firm.

Anchored in the two parallel underlying continuums of the internal-external focus and subjective-objective measurements, Figure 3 relates the quality dimensions and perspectives along a production life cycle. In order to investigate how quality dimensions and perspectives affect quality practices, a closer examination is needed. Built on Figure 3, a two-dimensional grid with internal-external focus and subjective-objective measurement as the reference

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axes is developed to create the maps of quality dimensions, perspectives and practices (see Figures 4, 5 and 6).

Quality dimensions (Figure 4) and quality perspectives (Figure 5) in the grid have already been explained in Figure 3. Similarly, in order to achieve the unified quality objective, quality management practices (Figure 5) have to accomplish the corresponding tasks along the dimensions from external to internal focus and from subjective to objective measurement. Customer-based practices, supplier-based practices, management-based practices, employee-based practices, information-based practices and product/process-based practices have to be aligned with quality dimensions and perspectives along a production cycle to achieve the competitive advantage by the superior quality.

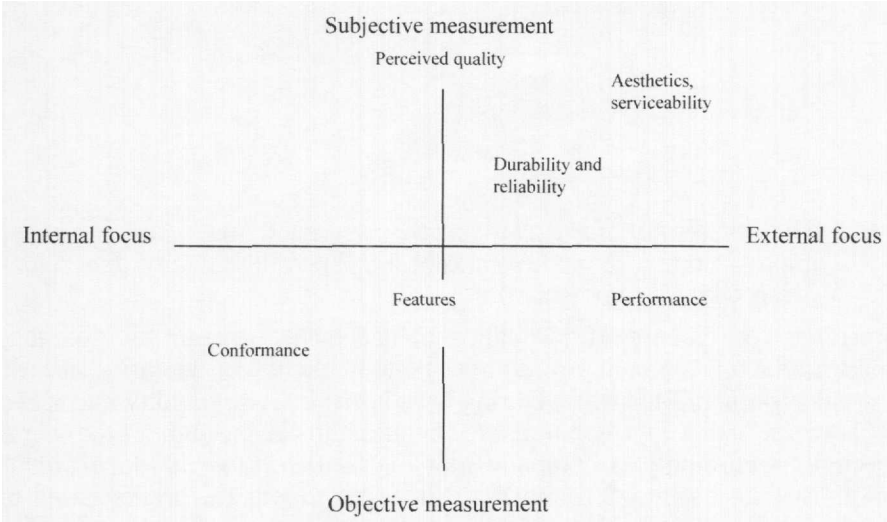


Figure 4.
Quality dimensions in a two-dimension grid

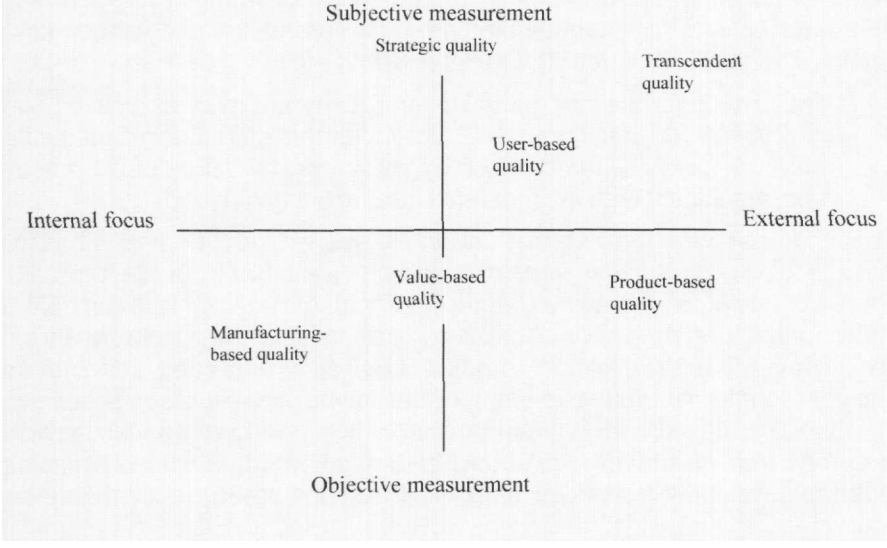


Figure 5.
Quality perspectives in a two-dimension grid

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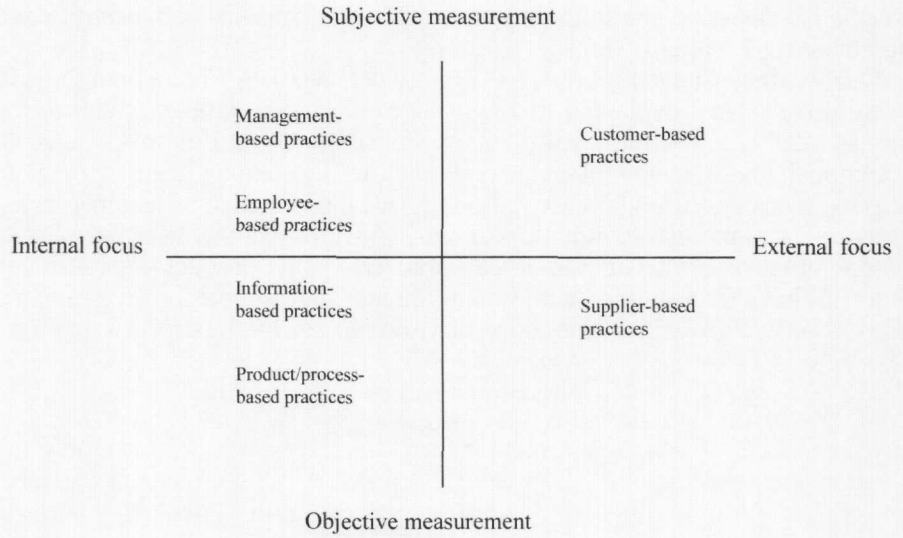


Figure 6.
Total quality management practices in a two-dimension grid

P3: The successful firms are more associated with an integrated implementation of quality concept (i.e. integrated quality dimensions, perspectives, and practices).

Sometimes, the failure of a quality management program is due to a concentration on internal procedures alone without consideration for the external customer. The manufacturing-based perspective of quality may direct managers' attention to only internal techniques, thus losing sight of the overall objective. Accordingly, a company that makes defect-free products cannot guarantee that it can sell the product. In order to win the orders based on superior quality, the firms should have an integrated view of quality including integrated dimensions, perspectives, and practices. Therefore, everyone along the production cycle has a unified interpretation of quality, and further, has a common goal stimulating the whole firm to achieve it.

P4: The firms with the external focus on quality are more associated with subjective measurements, while those with the internal focus on quality are more associated with objective measures; the successful firms are more associated with an integrated measure of quality.

Usually, firms with the external focus are more likely to use subjective measures of quality such as aesthetics and serviceability of product, product features, or customer satisfaction, while firms with the internal focus on quality emphasize more on the objective measures such as defect rate, conformance to specification, or performance of product. Logically connecting key external indicators to internal indicators is important for business success. If business objectives are aligned with internal indicators (e.g. customer satisfaction with defect rate), a cause-and-effect relationship should be studied. Such relationship is vital to developing a strategic objective (Reed and Montgomery, 1996). For

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example, Fuji-Xerox uses a variety of mathematical models relating such factors as copy quality, machine malfunctions, and maintenance time to customer satisfaction results.

P5: The path to achieving the integration in quality is by implementing and accumulating quality practices along with corresponding quality perspectives and dimensions; the internal focus on quality is the precondition to achieving the external focus.

The path to cultivate the integrated view of quality is either from the internal to the external focus or from the external to the internal focus. But, for either case, the appropriate internal procedures are necessary to achieve desired external results. If the firms only focus on the external measures such as good features to please the customers and they have no good conformance to the specification, they cannot have a steady source of customer or loyalty customers. If the firms have a solid competence to manufacture the product with a good conformance to the specifications and have eyes open to the outside customer requirements, then firms are more likely to succeed. Therefore, although we argue that an integrated view of quality is needed to achieve superior quality, the internal focus is a precondition.

Implications and conclusions

Sometimes managers understand well customer needs but still fail to gear all production and design systems toward that end. Coordinating all functions toward the final objective is essential to the success of TQM implementation (Ahire *et al.*, 1996; Babber, 1991). Failure is doomed if a design is not geared toward customer needs, or if the production is not geared toward meeting design specifications. TQM practices require the cooperation across all functions, even though each department has a different understanding about quality and quality practices. The two-dimensional grid (internal-external focus and subjective-objective measures) facilitates managers to coordinate activities of all departments and guide a firm to success.

Sharing information throughout the organization is absolutely critical for an organization to survive in today's competitive business world. A supply of consistent, accurate and timely information across all functional areas facilitates a better organizational response to rapidly changing customer needs. With a comprehensive information strategy, organizations can work with customers and suppliers to reduce cost, improve products and processes, and shorten cycle times (Buzzell and Wiersma, 1981). Daft (1998) discusses the importance of fitting the amount of information and its richness to organizational tasks and levels. For example, new product development requires coordinated information from marketing, design and engineering, manufacturing, and sales.

However, different measurement data have different subjective-objective degrees. Daft (1998) discussed the importance of fitting the amount of information and its richness to organizational tasks and levels. He defines information richness as the information-carrying capacity of data. The

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following media have decreasing degrees of richness: face-to-face, telephone and voice mail, written and addressed documents, written and impersonally addressed documents (Daft and Lengel, 1986). Therefore, it is helpful for managers to control and coordinate different kinds of quality practices with subjective-objective measures. With subjective measurement, managers can communicate effectively to resolve the equivocality among the quality definitions and practices. With objective measurement, managers can add more information to reduce uncertainty using some statistical process control tools.

In summary, strategic success depends upon a firm's ability to meet customer needs. However, a lot of firms implement TQM programs only internally. The internally implemented program must lead to the externally-oriented end. Yet, the external user-based and value-based perspectives alone cannot successfully guide the internal design and production functions in their quest to improve product or service quality. Therefore, companies should design services and products that meet customer needs. Designers need to use product-based perspectives of quality to describe product or service characteristics. So the product-based perspective of quality should reflect customers' definition of quality. Therefore, internal practices (management-based practices, employee-based practices, information-based practices, product/process-based practices) should consistently be combined with external practices (customer-based practices and supplier-based practices). For the practices with more objective measurement (information-based practices, product/process-based practices and supplier-based practices), managers need to emphasize how to add more information to make management rationalized. For the practices with more subjective measurements (management-based practices, employee-based practices and customer-based practices), managers need to emphasize how to communicate effectively to make people co-ordinated.

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